

Notes: Guiso, Sapienza, and Zingales (JF, 2008)

Trusting the Stock Market

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1 Big picture

- **Question.** Can *trust* explain why so many households stay out of the stock market, even though the equity premium is positive and even wealthy households often do not participate?
- **Core idea.** Buying stock is not just choosing a risky asset; it is also trusting that one will not be cheated by firms, managers, brokers, or other intermediaries. The paper defines trust as the subjective probability of *not* being cheated in financial transactions.
- **Setting.** The paper combines:
 - a simple portfolio model with “cheating risk,”
 - Dutch household survey data with direct trust questions and specially designed risk/ambiguity measures,
 - Italian bank-customer data on *personalized* trust in one’s own banker,
 - and cross-country evidence on stock market participation and ownership concentration.
- **Main theoretical result.** Lack of trust can generate nonparticipation even *without* fixed participation costs. Once fixed participation costs are added, trust amplifies them strongly: even a small perceived probability of being cheated can raise the wealth threshold for entering the stock market by a large multiple.
- **Main empirical result.** In Dutch micro data, trusting individuals are significantly more likely to own stocks and risky assets and, conditional on participating, they invest a larger share of wealth in them.
- **Main quantitative micro result.** Trusting others raises direct stock market participation by about 6.5 percentage points, roughly a 50% increase relative to the unconditional participation rate. It also raises the share of financial wealth held in stocks by about 3.4 percentage points.
- **Main “not just risk aversion” result.** The effect of trust survives controls for risk aversion, ambiguity aversion, optimism, and stock market expectations. Trust also predicts *more* stock diversification and *more* insurance demand, which is hard to reconcile with the view that trust is merely a proxy for low risk aversion or loss aversion.
- **Main personalized-trust result.** In the Italian bank-customer sample, people who trust their banker a lot are much more likely to invest in risky assets and to allocate more wealth to them.
- **Main cross-country result.** Countries with higher generalized trust have higher stock market participation and less concentrated ownership. The effect remains after controlling for legal enforcement and common-law origin.
- **Why this matters.** The paper provides a cultural / institutional explanation for the limited participation puzzle that is distinct from fixed costs, pure risk aversion, or ambiguity aversion.
- **Economic takeaway.** Trust acts like a wedge between the supply of financial claims and households’ willingness to hold them. If investors believe the game may be rigged, they may rationally stay out of the market.

2 Data and measurement (key objects)

2.1 Main Dutch household data

- **Primary micro data.** The core evidence comes from the 2003 wave of the Dutch National Bank (DNB) Household Survey.
- **Sample.** The survey contains detailed financial information for about 1,943 Dutch households (roughly 4,000 individuals). The special trust questionnaire was answered by about half the panel, producing a working sample of about 1,444 households / 1,990 individuals.
- **What is observed.**
 - Direct and indirect stockholding,
 - broader risky-asset participation,
 - portfolio shares in stocks and risky assets,
 - number of individual stocks held,
 - household financial wealth and income,
 - demographics,
 - private health insurance holdings,
 - and specially designed questions on trust, risk aversion, ambiguity aversion, optimism, and stock-market expectations.
- **Why the DNB data are valuable.** Most household-finance data sets do not jointly observe portfolio holdings and direct measures of trust, risk aversion, ambiguity aversion, and optimism. This survey does.

2.2 Measurement of generalized trust

- **Trust question.** The paper uses the standard World Values Survey question:

“Generally speaking, would you say that most people can be trusted or that you have to be very careful in dealing with people?”
- **Definition.** Trust is a dummy equal to one if the respondent answers “most people can be trusted.”
- **Empirical fact.** In the Dutch sample, trust is roughly one-third of respondents. The paper emphasizes that trust rises only mildly with wealth, so mistrust remains common even among the rich.
- **Why this matters.** If mistrust disappeared among wealthy households, it could not explain why many wealthy households still stay out of the stock market.

2.3 Measurement of risk aversion, ambiguity aversion, optimism, and expectations

- **Risk aversion.** Respondents report their willingness to pay for a risky lottery with known odds:

- 50 red balls, 50 black balls,
- payoff of 5,000 euros if red is drawn, zero otherwise.

The authors fit a CARA utility specification to map this willingness to pay into an Arrow–Pratt-style coefficient of absolute risk aversion.

- **Ambiguity aversion.** Respondents also price a lottery with uncertain urn composition, keeping expected value and variance similar to the risky lottery but introducing ambiguity in the probability distribution.
- **Optimism.** Measured using a Life Orientation Test style statement: whether the respondent expects more good things than bad things to happen.
- **Stock market expectations.** In a smaller subsample, respondents report whether they expect stock prices to rise over the next two years.
- **Key empirical point.** Trust is not strongly correlated with the direct measures of risk and ambiguity aversion, although it is positively correlated with optimism.

2.4 Key Dutch summary statistics

- Direct stockholders: about 13.5% of households.
- Holders of risky assets more broadly: about 42.2%.
- Portfolio share in stocks: about 3.3% in the full sample, but about 20.3% among stockholders.
- Portfolio share in risky assets: about 12.4% in the full sample, but about 29.5% among risky-asset holders.
- Number of stocks: average 0.53 in the full sample and 3.9 among stockholders.
- Private health insurance holders: about 26.9%.

Interpretation. The combination of low direct participation and meaningful variation in trust is exactly what the trust hypothesis needs.

2.5 Italian bank-customer data: personalized trust

- **Second micro data set.** The paper uses a survey of 1,834 customers of a large Italian commercial bank.
- **Why a second data set?** The Dutch data identify *generalized trust*. The Italian bank survey identifies *personalized trust* in the intermediary actually helping the investor.
- **Trust question.** Customers are asked:

“How much do you trust your bank official or broker as financial advisor for your investment decisions?”
- **Measure.** The paper creates high-trust and medium-trust dummies relative to low-trust customers.
- **Useful feature.** The survey also asks a direct qualitative question about attitudes toward risk, which helps address the concern that trust is just badly measured risk tolerance.

2.6 Cross-country data

- **Stock market participation across countries.** From Giannetti and Koskinen.
- **Ownership concentration and legal environment.** From La Porta et al.
- **Trust.** From the World Values Survey, measured as the fraction of people in each country who say most people can be trusted.
- **Cross-country coverage.** Participation ranges from roughly 2% in Turkey to around 40% in Australia.

Interpretation. These data allow the authors to move from “trust affects individual portfolios” to “trust shapes national financial development.”

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline trust model without participation costs

The paper starts with a two-asset portfolio problem. A household allocates wealth W between:

- a safe asset yielding return r_f ,
- and a risky stock with true return $\tilde{r} \sim N(\bar{r}, \sigma^2)$.

In addition to ordinary risk, the stock is exposed to a *cheating* probability p : with probability p , the value of the investment vanishes. This captures fraud, expropriation, or intermediary theft. Trust is therefore $1 - p$.

The investor chooses the portfolio share α in stocks to solve

$$\max_{\alpha} (1 - p)\mathbb{E}U(\alpha\tilde{r}W + (1 - \alpha)r_fW) + pU((1 - \alpha)r_fW). \quad (1)$$

The first-order condition is

$$(1 - p)\mathbb{E}U'(\alpha\tilde{r}W + (1 - \alpha)r_fW) (\tilde{r} - r_f) \leq pU'((1 - \alpha)r_fW) r_f. \quad (1)$$

Interpretation.

- The left-hand side is the expected marginal benefit of moving one more dollar into stock.
- The right-hand side is the expected marginal cost of complete loss if cheating occurs.
- If the cost dominates at $\alpha = 0$, the household stays out of the stock market.

3.2 Participation threshold

From equation (1), the investor participates only if the subjective cheating probability is below

$$\bar{p} = \frac{\bar{r} - r_f}{\bar{r}}. \quad (2)$$

So:

Proposition 1. Only investors with high enough trust, that is $(1 - p) > (1 - \bar{p})$, invest in the stock market.

Interpretation.

- Trust works by lowering the subjective expected return on equity.
- If mistrust is strong enough, the risky asset is not just unattractive — it is dominated by the safe asset.
- The paper calibrates this threshold using postwar U.S. numbers and gets a nonparticipation threshold around 6.25% cheating probability in the no-participation-cost case.

3.3 Conditional portfolio share

If $p < \bar{p}$, the investor participates and chooses an interior share $\alpha^* > 0$. The paper shows:

Proposition 2. The more an investor trusts, the higher his optimal portfolio share invested in stocks conditional on participation.

Under exponential utility and normal returns, the optimal stock share can be written as a standard mean–variance term minus an additional penalty term that is increasing in p . The intuition is enough:

- with no cheating risk, the investor holds the usual risky-asset share,
- as cheating risk rises, the optimal share in stocks falls.

Interpretation. Trust affects both the extensive margin (whether to participate) and the intensive margin (how much to invest once participating).

3.4 Adding participation costs

The paper then introduces a fixed participation cost f . If the investor enters the stock market, he pays f upfront and allocates only the remaining wealth $W - f$ between safe and risky assets.

The participation condition becomes

$$(1 - p)\mathbb{E}U\left(\alpha_p^* \tilde{r}(W - f) + (1 - \alpha_p^*)r_f(W - f)\right) + pU\left((1 - \alpha_p^*)r_f(W - f)\right) > U(r_f W). \quad (3)$$

This leads to two important results:

Proposition 3. Adding a fixed participation cost lowers the threshold value of p that triggers nonparticipation.

Proposition 4. For any probability of being cheated p , there exists a wealth threshold $\bar{W}_p$ that triggers participation, and $\bar{W}_p$ is increasing in p .

Interpretation.

- Trust and participation costs reinforce one another.
- Fixed costs make mistrust more damaging, because the investor must pay the cost before knowing whether the investment is safe from cheating.
- Mistrust, in turn, raises the effective wealth threshold needed to justify paying the fixed cost.

This is the paper's key amplification result.

3.5 Calibration of participation costs: Table II

The paper calibrates the model with CRRA utility, relative risk aversion 3, return on equity 1.12, risk-free rate 1.05, variance of stock returns 0.05, and a participation cost equal to 1.6% of the threshold wealth level under full trust.

Table II shows:

- if cheating probability is 0.5%, the wealth threshold rises by about 45%;
- if cheating probability is 1%, the wealth threshold roughly doubles;
- if cheating probability is 2%, the wealth threshold rises about fivefold;
- the mistrust threshold for nonparticipation falls from about 6.25% without participation costs to about 3% once fixed costs are introduced.

Interpretation. Tiny levels of mistrust become economically powerful once participation costs are present.

3.6 Diversification: trust versus risk aversion

A central concern is that trust might just proxy for risk tolerance. The paper therefore derives a prediction in which trust and risk tolerance move in different directions: the *number of stocks held*.

In the two-stock case, if there is a per-stock cost c , buying the second stock is worthwhile if

$$(1 - p)[D + pV] > c, \quad (4)$$

where

$$D = \mathbb{E}U\left(\frac{1}{2}\tilde{W}_1 + \frac{1}{2}\tilde{W}_2\right) - \mathbb{E}U(\tilde{W}_1), \quad (4)$$

$$V = \left[\mathbb{E}U\left(\frac{1}{2}\tilde{W}_1\right) + \mathbb{E}U\left(\frac{1}{2}\tilde{W}_2\right)\right] - \left[\mathbb{E}U\left(\frac{1}{2}\tilde{W}_1 + \frac{1}{2}\tilde{W}_2\right) + U(0)\right]. \quad (5)$$

Interpretation.

- D is the standard diversification benefit from reducing idiosyncratic risk.
- V is the additional benefit from diversifying *cheating risk*.

This yields:

Proposition 5. Diversification is nondecreasing in trust if $D > V$.

Proposition 6. Diversification is nondecreasing in trust if the standard diversification benefit already exceeds the per-stock cost in the no-trust-problem benchmark.

The paper then generalizes to n stocks with exponential utility:

$$\max_n \sum_{g=0}^n C_n^g p^g (1-p)^{n-g} \mathbb{E} \left[-e^{-\theta(W/n) \sum_{i=1}^g \tilde{r}_i} \right] - cn. \quad (5)$$

Figure 1 solves this numerically and shows:

- more risk aversion leads to *more* stocks because diversification becomes more valuable;
- more trust also leads to more stocks;
- therefore trust and *risk tolerance* have opposite implications for the number of stocks.

Interpretation. If trust predicts the number of stocks held, that is not what one would expect if trust were merely a stand-in for low risk aversion.

3.7 Main empirical specifications

The paper then takes the model to the data using reduced-form regressions that mirror the model's margins.

Direct stock market participation.

$$\Pr(\text{DirectStock}_i = 1) = \Phi(\beta_0 + \beta_1 \text{Trust}_i + \beta_2 \text{RiskAversion}_i + \beta_3 \text{AmbiguityAversion}_i + X_i' \gamma). \quad (6)$$

Participation in risky assets.

$$\Pr(\text{RiskyAssetHolder}_i = 1) = \Phi(\beta_0 + \beta_1 \text{Trust}_i + \beta_2 \text{RiskAversion}_i + \beta_3 \text{AmbiguityAversion}_i + X_i' \gamma). \quad (7)$$

Portfolio shares. For portfolio shares in stocks and risky assets, the paper uses Tobit specifications because many households choose zero holdings.

Number of stocks. For the number of distinct stocks held, the paper uses Poisson count regressions.

Health insurance demand. To distinguish trust from risk aversion, the paper also runs probit/Tobit regressions of private health-insurance demand on trust and the same controls.

Interpretation.

- The model predicts extensive-margin, intensive-margin, and diversification effects.
- The regressions are directly organized around those three objects.

3.8 Cross-country specifications

At the country level, the paper estimates regressions of the form

$$\text{StockParticipation}_c = \alpha + \beta \text{Trust}_c + \gamma \text{LegalEnforcement}_c + \delta \text{CommonLaw}_c + \varepsilon_c, \quad (8)$$

$$\text{CloselyHeld}_c = \alpha + \beta \text{Trust}_c + \gamma \text{LegalEnforcement}_c + \delta \text{CommonLaw}_c + \varepsilon_c. \quad (9)$$

Interpretation.

- If trust matters for the demand for equity, then countries with higher trust should have:
 - more households owning stock,
 - and more diffuse ownership structures.
- This is exactly what the cross-country evidence tests.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the trust effect in the Dutch micro data

The main source of identification is cross-household variation in trust after controlling for:

- financial wealth,
- income,
- age, gender, education, household size, and number of children,
- measured risk aversion,
- measured ambiguity aversion,
- optimism,
- and, in a smaller subsample, expected stock market performance.

Interpretation. The paper asks whether two households with similar wealth, demographics, and measured attitudes toward risk differ in participation because one trusts more than the other.

4.2 Why trust is not just risk aversion

This is one of the paper's strongest parts.

- **Direct controls.** Trust remains important after controlling for measured risk aversion and ambiguity aversion.
- **Diversification test.** Trust predicts holding more stocks; this is not what one should see if trust is simply low risk aversion.
- **Insurance test.** Trust is positively related to private health-insurance demand, whereas low risk aversion should predict the opposite.

- **Italian data.** In the Italian bank-customer sample, even with a more direct measure of risk attitude, trust in the banker still matters.

Interpretation. The paper does not just say “we controlled for risk aversion.” It derives margins on which trust and risk aversion should differ, and then tests those margins.

4.3 Why trust is not just optimism or bullish expectations

- The authors include a direct optimism measure.
- They also include a direct measure of expected stock market appreciation in the smaller subsample where it is available.
- In both cases, the trust coefficient remains similar in magnitude.

Interpretation. Trust is not merely “people who think returns will be high.”

4.4 Why education matters for credibility

The model interprets trust as a prior. If so, people with more education or market knowledge should rely less on this prior because they have more other information.

That is exactly what the paper finds:

- trust matters more for lower-education households,
- and its effect weakens among more educated households.

Interpretation. This is a useful consistency check for the theory, because it is hard to explain by a simple omitted-preference story.

4.5 What makes the personalized-trust evidence valuable

The Italian bank data strengthen the argument because:

- they use trust in a specific intermediary rather than generalized trust,
- the respondents are already bank customers, so the question is directly about the financial intermediary through which many stock purchases occur,
- and the survey includes a more direct risk-attitude measure.

Interpretation. If both generalized trust and trust in one’s own banker predict risky-asset demand, the trust channel is harder to dismiss.

4.6 Cross-country credibility

The cross-country exercise is more suggestive than the household-level evidence, but it is still interesting because:

- trust predicts participation even after controlling for legal enforcement and common-law origin;

- trust predicts lower ownership concentration;
- and the trust effect is strongest in countries whose return histories were disrupted by World War II occupation, which is consistent with the model's idea that trust matters more when investors cannot rely on long stationary histories of returns.

4.7 Main limitations

- Trust is still a survey measure, so measurement error and interpretation issues remain.
- Risk and ambiguity aversion are measured from hypothetical lotteries, which are noisy.
- The cross-country evidence is low-dimensional and potentially subject to many omitted institutional variables.
- The paper is strongest on showing that trust matters for participation, but weaker on pinning down where trust itself comes from and how quickly it changes.

4.8 Relation to the literature

- Relative to fixed-cost explanations of stock market nonparticipation, the paper shows that trust can generate nonparticipation even without costs and powerfully amplifies costs when they exist.
- Relative to ambiguity-aversion explanations, the paper argues that trust is conceptually different and empirically separable.
- Relative to behavioral explanations based on optimism or expectations, it shows trust has independent predictive power.
- Relative to the culture-and-finance literature, it gives a concrete portfolio-choice mechanism through which culture can affect financial outcomes.

5 Tables and figures

5.1 Table I and Figure 2: stock market participation across countries

Read:

- Table I is a motivating table, not the main causal evidence. It reports participation in risky assets across wealth quartiles for a set of countries.
- The cross-country dispersion is enormous. For direct stockholding, the top 5% wealth group participates at about 80.6% in Sweden but only 30.8% in Italy, 25.7% in Austria, and 14.4% in Spain.
- Once indirect participation is included, the U.S. top 5% reaches 93.7%, Sweden 95.8%, but Spain only 24.6% and Greece 32.9%.
- Figure 2 compresses the same point visually: direct stock market participation ranges from roughly 2% in Turkey to about 40% in Australia.

Table I
**Proportion of Households Investing in Risky Assets,
by Asset Quartiles**

Panel A shows the proportion of households in each quartile of gross financial wealth that own stock directly. Panel B shows the same proportion when we include also indirect ownership, via mutual funds or pension funds. Data for European countries are computed from the 2004 wave of the Survey for Health, Age, and Retirement in Europe (Share), and refer to year 2003. Data for the U. S. are drawn from the 1998 Survey of Consumer Finances. Data for the U. K. are drawn from the 1997–1998 Financial Research Survey.

	Quartile I	Quartile II	Quartile III	Quartile IV	Top 5%	Average
Panel A. Direct Stockholding						
U.S.	1.4	6.9	20.6	47.9	70.1	19.2
U.K.	0.0	4.4	28.3	53.6	67.9	21.6
Netherlands	1.5	7.4	20.0	40.3	60.2	17.2
Germany	0.6	4.1	16.1	36.1	50.5	14.0
Italy	0.0	0.8	3.1	12.8	30.8	4.0
Austria	0	1.7	2.8	15.6	25.7	5.0
Sweden	12.9	30.7	46.9	72.8	80.6	40.8
Spain	0	0.3	1.8	13.2	14.4	3.5
France	0.7	9.9	14.6	33.3	44.2	14.4
Denmark	6.3	25.9	36.4	55.6	68.4	31.0
Greece	0	0.7	3.2	17.3	23.5	4.9
Switzerland	2.8	12.2	30.3	54.2	63.2	24.9
Panel B. Direct and Indirect Stockholding						
U.S.	4.4	38.3	66.0	86.7	93.7	48.9
U.K.	4.9	11.9	37.8	71.1	83.9	31.5
Netherlands	1.7	11.0	31.3	52.8	72.0	24.1
Germany	1.5	11.8	28.7	51.4	61.2	22.9
Italy	0.0	0.8	5.2	27.5	64.8	8.2
Austria	0	1.9	8.1	25.5	33.8	8.8
Sweden	25.8	63.4	82.7	92.9	95.8	66.2
Spain	0	1.1	3.0	19.1	24.6	5.4
France	1.1	17.6	29.9	57.6	67.3	26.2
Denmark	6.6	30.8	44.8	65.7	75.4	37.0
Greece	0	0.7	4.0	22.2	32.9	6.3
Switzerland	2.8	20.0	38.2	63.7	65.8	31.4

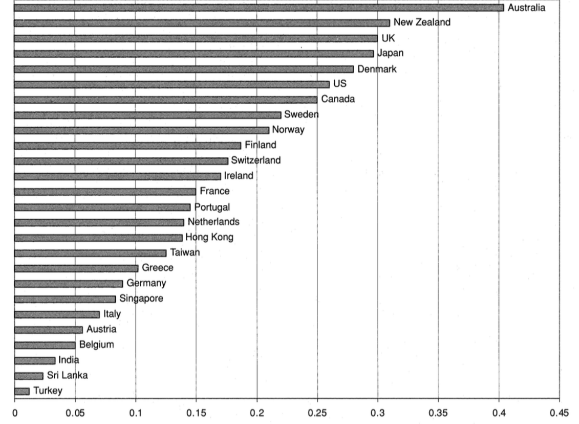


Figure 2. Stock market participation across countries. The figure shows the fraction of individuals that invest directly in stocks across countries. The data come from Giannetti and Koskinen (2005).

Why it matters. The paper needs to explain not just why some households abstain, but why abstention is so widespread and why it varies so much across countries and wealth groups.

5.2 Table II and Figure 1: theory calibration and diversification

Table II
Limited Trust and Participation Costs

This table shows the result of a calibration exercise of the optimal portfolio choice for different levels of trust (expressed as the perceived probability p that an investor will be cheated). The first column reports these perceived probabilities, varying between zero and the maximum value above which no participation takes place. Column 2 reports the wealth threshold beyond which people invest in the stock market expressed as a ratio of the level of wealth that will trigger investment in the absence of trust considerations. Column 3 reports the optimal portfolio share invested in the stock market, conditional on investing. The calculations assume the investor has CRRA utility; relative risk aversion is three; the level of initial wealth is one; participation cost is 1.6% of the wealth threshold that leads to nonparticipation when the probability of being cheated is equal to zero; and the return on equity and the risk-free rate are 1.12 and 1.05, respectively, while the variance of stock returns is 0.05.

Probability of Being Cheated and Wealth Participation Threshold		
Probability of Being Cheated in the Stock Market	Wealth Participation Thresholds/Wealth Threshold When Trust Is Full ($p = 0$)	Optimal Share Invested in Stocks if Participation Occurs
0	1.000	0.519
0.001	1.096	0.463
0.005	1.458	0.356
0.01	1.980	0.285
0.0125	2.294	0.258
0.0150	2.657	0.235
0.0175	3.585	0.195
0.020	4.920	0.152
0.0225	6.942	0.133
0.03		0.000

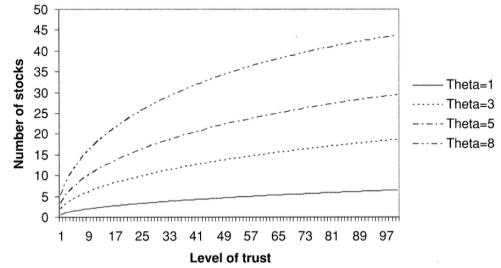


Figure 1. Trust and the number of stocks. This figure reports the optimal number of stocks an investor should hold for different levels of risk aversion and different levels of trust. Trust is the percentage probability of being cheated. Theta is the coefficient of absolute risk aversion of an exponential utility function.

Read:

- Table II quantifies the amplification role of trust when participation is costly.
- The headline numbers are memorable:
 - cheating probability 0.5%: participation wealth threshold up about 45%,
 - cheating probability 1%: threshold roughly doubles,

– cheating probability 2%: threshold roughly quintuples.

- Figure 1 shows that the optimal number of stocks rises with trust for every risk-aversion level shown.

Why it matters. This is the paper’s main bridge from theory to empirics: trust should affect not only whether investors enter, but also how diversified they are once they enter.

5.3 Table III: summary statistics

Table III
Summary Statistics

The table shows summary statistics of the variables used. Panels A–E use data from the Dutch National Bank survey. Financial wealth, income, and health insurance premiums are in thousand euros. Trust is a dummy equal to one if a person answers “most people can be trusted” to the question: “Generally speaking, would you say that most people can be trusted or that you have to be very careful in dealing with people?” The price for the lotteries is obtained by asking people how much they are willing to pay to participate in a lottery. In the unambiguous lottery the interviewee is given the exact number of balls in the urn. In the ambiguous one this number is uncertain, but the interviewee is given the probability distribution. The coefficient of risk aversion is obtained fitting a CARA utility. Optimism is an index of agreement (from 1 to 5) to the statement “I expect more good things to happen to me than bad things.” “Expect stock market to go up” is a dummy equal to one if the interviewee answers “increase” to the question “do you expect market stock prices to increase, remain constant or decrease in the next two years?” Panel F is from a survey of bank customers of a large Italian commercial bank. In this sample high trust is a dummy equal to one when a bank customer responds “a lot” or “enough” to the question: “How much do you trust your bank official or broker as financial advisor for your investment decisions?” Medium trust is a dummy variable equal to one if s/he answers “so so” or “not much” (the omitted category is “not at all”). Panel G is from an international data set combining data from Giannetti and Koskinen (2005), La Porta et al. (1998), and the World Values Survey (data on trust).

	Mean	Median	SD	Min	Max
Panel A. Stock Holdings, Financial Assets and Income: DNB (N = 1,444)					
Direct stockholders	0.135	0.0	0.342	0	1
Risky assets holders	0.422	0.0	0.449	0	1
Portfolio share in stocks	0.033	0	0.119	0	1
Portfolio share in stocks among stockholders	0.203	0.118	0.229	0.0001	0.926
Portfolio share in risky assets	0.124	0	0.230	0	1
Portfolio share in risky assets among holders of risky assets	0.295	0.195	0.274	0.001	1
Number of stocks	0.532	0	2.873	0	97
Number of stocks among stockholders	3.90	3	6.952	1	97
Holders of health insurance	0.269	0	0.444	0	1
Health insurance premium ('000 of euros)	0.178	0	1.148	0	44.411
Household financial wealth ('000 of euros)	031.230	10.140	66.804	0	838.041
Gross household income ('000 of euros)	28.128	22.362	68.930	0	2,197.032
Number of observations					
Panel B. Trust, Risk Aversion, Ambiguity Aversion, and Optimism: DNB (N = 1,444)					
Trust WVS	0.332	0.0	0.471	0	1
Absolute risk aversion	0.107	0.028	0.186	0	0.693
Ambiguity aversion	4.155	7.1077	4.275	–2.389	41.692
Price to participate in risky lottery ('000 of euros)	0.123	0.001	0.421	0	5
Price to partic. in ambiguous lottery ('000 of euros)	0.090	0.001	0.341	0	3
Individual optimism	3.127	4	1.532	0	5
Expect stock market to go up	0.596	1	0.458	0	1

(continued)

Table III—Continued

Panel C. Correlation Matrix between Trust, Risk Aversion, Ambiguity Aversion, and Optimism: DNB (N = 1,444)							
	Trust WVS	Absolute risk Aversion	Ambiguity Aversion	Optimism			
Trust WVS	1						
Absolute risk aversion	0.017	1					
Ambiguity aversion	0.014	0.072	1				
Optimism	0.310	0.172	0.013	1			
Panel D. Trust and Wealth: DNB							
Financial Wealth							
	Quartile I	Quartile II	Quartile III	Quartile IV	Top 5%	Average	
Trust WVS	0.342	0.373	0.409	0.396	0.365	0.382	
			Mean	Median	SD	Min	Max
Panel E. Household Head and Household Demographics: DNB							
Male			0.466	0	0.499	0	1
Age			30.184	34	27.011	0	90
High education			0.178	0	0.382	0	1
Medium education			0.292	0	0.455	0	1
Employee			0.369	0	0.483	0	1
N. of household members			2.442	2	1.281	1	8
N. of children in the household			0.711	0	1.070	0	6
Panel F. Summary Statistics for the Bank Customer Dataset (N = 1,834)							
Share holding risky assets			0.638	1	0.481	0	1
Portfolio share of risky assets			0.223	0.110	0.269	0	1
High trust in bank official			0.665	1	0.472	0	1
Medium trust in bank official			0.135	0	0.342	0	1
Averse to risk			0.709	1	0.454	0	1
Financial assets ('000, euros)			109.185	25	270.810	0	3,760
Male			0.711	1	0.453	0	1
Age			54.698	56	14.366	21	85
Years of education			11.974	13	4.412	0	21
Panel G. Summary Statistics for the International Data (N = 33)							
% of stock market cap. closely held			44.03	42.43	18.38	7.94	77.48
% population participating in the stock market			0.16	0.15	0.10	0.01	0.40
Average trust			0.34	0.36	0.17	0.03	0.67
Legal enforcement			0.54	0.50	0.24	0.17	1.00
Common law			0.32	0.00	0.47	0.00	1.00

Read:

- Direct stockholding is rare (13.5%), while broader risky-asset holding is much more common (42.2%).
- Among stockholders, the average portfolio share in stocks is about 20.3%.
- Among risky-asset holders, the average risky-asset share is about 29.5%.
- The trust dummy is around one-third.
- Trust rises only mildly with wealth.

Why it matters. The summary statistics already suggest that trust has the potential to explain two puzzles at once:

- why many households own no stock at all,
- and why even rich households may still stay out.

5.4 Table IV and Table V: participation in stocks and risky assets

Table IV
The Effect of Trust on Direct Stock Market Participation

The dependent variable is a dummy equal to one if the household directly owns shares in a company (be it listed or not) except in his own company. The table reports the probit estimates, calculated as the effect on the LHS of a marginal change in the RHS variable computed at the average value of the RHS variables. All household characteristics, which are defined in Table I, are assumed to be those of the household head. Standard errors are reported in parentheses. *** indicates the coefficient is different from zero at the 1% level, ** at the 5% level, and * at the 10% level.

	Whole Sample				Above-median Wealth
	(1)	(2)	(3)	(4)	(5)
Trust	0.065*** (0.023)	0.059*** (0.022)	0.057*** (0.022)	0.064 (0.051)	0.072** (0.036)
Risk aversion	0.055 (0.052)	0.061 (0.047)	0.061 (0.047)	0.012 (0.122)	0.113 (0.085)
Ambiguity aversion		-0.002 (0.002)	-0.002 (0.002)	-0.001 (0.004)	-0.003 (0.003)
Optimism			0.005 (0.010)	0.047* (0.025)	0.023 (0.019)
Stock market expected to go up				-0.020 (0.043)	
Financial wealth	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001** (0.000)	0.001*** (0.000)
Income	0.994 (1.325)	0.837 (1.190)	0.824 (1.189)	-7.001 (20.720)	3.831 (3.662)
Male	0.039 (0.027)	0.036 (0.024)	0.036 (0.024)	0.025 (0.069)	0.047 (0.045)
Age	-0.005** (0.002)	-0.004* (0.002)	-0.005* (0.002)	-0.010* (0.005)	-0.006 (0.004)
Age squared	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000* (0.000)	0.000 (0.000)
Household size	-0.015 (0.026)	-0.014 (0.023)	-0.014 (0.023)	0.041 (0.060)	-0.075* (0.045)
Number of children	0.040 (0.030)	0.037 (0.028)	0.037 (0.028)	0.009 (0.065)	0.121** (0.054)
College education	0.072** (0.036)	0.066** (0.033)	0.063* (0.033)	0.357*** (0.133)	0.072 (0.053)
High school education	0.041 (0.029)	0.038 (0.027)	0.036 (0.027)	0.169* (0.091)	0.055 (0.047)
Employee	-0.002 (0.030)	-0.000 (0.027)	-0.002 (0.027)	-0.139** (0.067)	-0.058 (0.053)
Observations	1156	1156	1156	255	618

Table V
The Effect of Trust on Participation in Risky Assets

The dependent variable is a dummy equal to one if the household directly owns any risky asset (shares, mutual funds, corporate bonds, put and call options) except equity in one's own company. The table reports the probit estimates, calculated as the effect on the LHS of a marginal change in the RHS variable computed at the average value of the RHS variables. All household characteristics, which are defined in Table I, are assumed to be those of the household head. Standard errors are reported in parentheses. *** indicates the coefficient is different from zero at the 1% level, ** at the 5% level, and * at the 10% level.

	Whole Sample				Above-median Wealth
	(1)	(2)	(3)	(4)	(5)
Trust	0.085** (0.037)	0.084** (0.037)	0.082** (0.037)	0.053 (0.079)	0.084* (0.044)
Risk aversion	-0.100 (0.091)	-0.107 (0.092)	-0.106 (0.092)	-0.039 (0.197)	0.019 (0.115)
Ambiguity aversion		0.000 (0.000)	0.000 (0.000)	0.001 (0.006)	0.000 (0.000)
Optimism			0.007 (0.019)	-0.009 (0.040)	0.042* (0.023)
Stock market exp. to go up				-0.028 (0.077)	
Financial wealth	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
Income	-1.951 (3.271)	-1.979 (3.290)	-2.006 (3.295)	-53.158 (33.834)	-4.451 (5.356)
Male	0.109** (0.048)	0.109** (0.048)	0.109** (0.048)	0.153 (0.116)	0.096 (0.062)
Age	-0.007 (0.004)	-0.006 (0.004)	-0.007 (0.005)	-0.009 (0.009)	-0.011* (0.006)
Age squared	0.000* (0.000)	0.000* (0.000)	0.000* (0.000)	0.000 (0.000)	0.000 (0.000)
Household size	0.165*** (0.044)	0.165*** (0.044)	0.164*** (0.044)	0.083 (0.096)	0.119** (0.056)
Number of children	-0.109** (0.052)	-0.109** (0.052)	-0.108** (0.052)	0.023 (0.108)	-0.051 (0.066)
College education	0.016 (0.050)	0.016 (0.050)	0.013 (0.051)	0.136 (0.111)	-0.029 (0.061)
High school education	0.020 (0.045)	0.019 (0.045)	0.017 (0.045)	0.083 (0.094)	0.011 (0.057)
Employee	0.183*** (0.050)	0.183*** (0.050)	0.181*** (0.050)	0.203* (0.107)	0.109* (0.066)
Observations	1,007	1,007	1,007	237	618

Read:

- Table IV studies *direct* stock market participation.
- The benchmark trust coefficient is about 0.065, meaning trust raises direct stockholding by 6.5 percentage points.
- Since the unconditional direct participation rate is only about 13.5%, this is roughly a 50% increase.
- The coefficient survives controls for risk aversion, ambiguity aversion, optimism, and stock-market expectations.
- The effect remains present, and even somewhat larger, in the above-median-wealth subsample.

Table V:

- The same exercise for broader risky-asset participation.
- Trust raises risky-asset participation by about 8.5 percentage points, around 20% of the sample mean.

Why it matters. These two tables establish the paper’s baseline fact: trusting households participate more.

5.5 Table VI and Table VII: portfolio shares and education heterogeneity

Table VI
The Effect of Trust on the Portfolio Share in Stocks and Risky Assets

The table reports Tobit estimates for the portfolio share invested in stocks (Panel A) and in risky assets (Panel B), except equity in one’s own company. The share in stocks is the value of household holdings of listed and unlisted stocks divided by total household financial assets; the share in risky assets is the value in stocks, stock mutual funds, corporate bonds, and put and call options divided by total family financial wealth. All characteristics are those of the household head. Standard errors are reported in parentheses. *** indicates the coefficient is different from zero at the 1% level, ** at the 5% level, and * at the 10% level.

Panel A. Share of Household in Financial Assets Invested in Stocks				
Trust	0.131*** (0.047)	0.133*** (0.047)	0.130*** (0.048)	0.145 (0.119)
Risk aversion	0.064 (0.116)	0.085 (0.118)	0.085 (0.118)	−0.048 (0.332)
Ambiguity aversion	−0.003 (0.003)	−0.003 (0.003)	0.003 (0.010)	0.003 (0.010)
Optimism			0.012 (0.026)	0.088 (0.068)
Stock market expected to go up				−0.039 (0.119)
Financial wealth	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.001 (0.001)
Income	0.873 (3.293)	0.781 (3.274)	0.754 (3.274)	−19.128 (53.827)
Male	0.129* (0.067)	0.132* (0.067)	0.132* (0.067)	0.161 (0.211)
Age	−0.011** (0.005)	−0.011** (0.005)	−0.012** (0.006)	−0.028** (0.014)
Age squared	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)
Household size	−0.066 (0.058)	−0.067 (0.058)	−0.067 (0.058)	0.071 (0.166)
Number of children	0.139** (0.068)	0.141** (0.068)	0.141** (0.068)	0.141 (0.179)
College education	0.169** (0.067)	0.171** (0.067)	0.166** (0.068)	0.724*** (0.232)
High school education	0.087 (0.063)	0.089 (0.063)	0.086 (0.064)	0.400* (0.217)
Employee	−0.012 (0.067)	−0.010 (0.067)	−0.014 (0.067)	−0.348** (0.170)
Observations	999	999	999	234

(continued)

Table VI—Continued

Panel B. Share of Household Financial Assets Invested in Risky Financial Assets				
Trust	0.096*** (0.034)	0.096*** (0.034)	0.095*** (0.035)	0.022 (0.071)
Risk aversion	−0.093 (0.086)	−0.095 (0.087)	−0.095 (0.087)	−0.068 (0.181)
Ambiguity aversion		0.000 (0.000)	0.000 (0.000)	0.004 (0.005)
Optimism			0.004 (0.018)	−0.003 (0.035)
Stock market expected to go up				−0.030 (0.070)
Financial wealth	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.001* (0.000)
Income	−2.152 (3.295)	−2.164 (3.301)	−2.180 (3.304)	−32.009 (30.996)
Male	0.125*** (0.048)	0.124*** (0.048)	0.125*** (0.048)	0.231** (0.117)
Age	−0.009** (0.004)	−0.009** (0.004)	−0.010** (0.004)	−0.014* (0.008)
Age squared	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)
Household size	0.066 (0.043)	0.066 (0.043)	0.065 (0.043)	−0.034 (0.095)
Number of children	−0.007 (0.049)	−0.007 (0.049)	−0.007 (0.049)	0.169 (0.104)
College education	0.028 (0.047)	0.028 (0.047)	0.027 (0.048)	0.159 (0.101)
High school education	0.017 (0.042)	0.017 (0.042)	0.015 (0.043)	0.009 (0.086)
Employee	0.120** (0.049)	0.120** (0.049)	0.119** (0.049)	0.096 (0.101)
Observations	999	999	999	234

Read:

- Table VI moves from the extensive margin to the intensive margin.
- Trust increases the portfolio share invested in stocks by about 0.131 in the Tobit specification, which the paper translates into about a 3.4 percentage point increase in the stock share.
- For risky assets more broadly, the trust coefficient is about 0.096.

Table VII:

- Splits the sample by education.
- For lower-education households, the trust effect on direct stock ownership is about 0.059; for higher-education households it drops to about 0.014 and is not statistically different from zero.
- The same pattern appears for broader risky-asset ownership and for portfolio shares.

Why it matters. These are exactly the patterns predicted by a model in which trust is a prior that matters more when information is scarce.

5.6 Table VIII: trust is not a proxy for risk aversion

Read:

Table VII
Trust and Education

In this table we re-estimate the regressions in Table IV (first two columns), Table V (second two columns), and Table VI (last four columns) splitting the sample by level of education. In the first two columns the left-hand side variable is a dummy equal to one if the household holds equity directly. In the second two columns is a dummy equal to one if the household holds stocks directly or indirectly and/or invests in corporate bonds and put and call options. In the remaining columns the left hand side variable is the share of household financial assets invested directly in equity (columns 5 and 6) and in risky assets (last two columns), respectively; the share in risky assets is the value in stocks, stock mutual funds, corporate bonds, and put and call options divided by total family financial wealth. In all cases, investment in stock does not include equity in one's own company. As in Tables IV and V, the first four columns are probit estimates, calculated as the effect on the LHS of a marginal change in the RHS variable computed at the average value of the RHS variables. The last four columns are Tobit estimates. High education includes all those with a high school degree or a university degree. Low education includes all those with less than high school education. Education is that of the head of the household. All characteristics are those of the household head. Standard errors are reported in parentheses. *** indicates the coefficient is different from zero at the 1% level, ** at the 5% level, and * at the 10% level.

	Ownership of Stock		Ownership of Risky Assets		Share of Stocks		Share of Risky Assets	
	Low Educ	High Educ	Low Educ	High Educ	Low Educ	High Educ	Low Educ	High Educ
Trust	0.059** (0.025)	0.014 (0.046)	0.095** (0.045)	0.056 (0.068)	0.155*** (0.052)	0.071 (0.095)	0.119*** (0.040)	0.052 (0.063)
Risk aversion	0.018 (0.038)	0.229* (0.118)	-0.094 (0.105)	-0.201 (0.195)	-0.004 (0.126)	0.288 (0.250)	-0.102 (0.097)	-0.174 (0.186)
Ambiguity aversion	-0.003*** (0.001)	-0.001 (0.002)	0.000 (0.000)	0.001 (0.002)	-0.007 (0.005)	-0.002 (0.003)	0.000 (0.000)	0.001 (0.001)
Optimism	-0.000 (0.008)	0.032 (0.029)	-0.001 (0.022)	0.021 (0.040)	-0.003 (0.026)	0.096 (0.061)	-0.006 (0.020)	0.021 (0.037)
Financial wealth	0.001*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.003*** (0.001)	0.002*** (0.000)	0.002*** (0.001)	0.002*** (0.000)	0.002*** (0.000)
Income	-0.127 (1.136)	3.785 (3.355)	-3.118 (5.310)	-0.618 (5.426)	-0.278 (3.653)	2.700 (7.012)	-3.006 (4.860)	-1.194 (5.336)
Male	0.022 (0.022)	0.068 (0.050)	0.141** (0.061)	0.021 (0.080)	0.089 (0.080)	0.216* (0.122)	0.132** (0.061)	0.069 (0.076)
Age	-0.002 (0.002)	-0.009 (0.011)	-0.005 (0.005)	-0.035* (0.018)	-0.004 (0.006)	-0.025 (0.023)	-0.003 (0.005)	-0.049*** (0.016)
Age squared	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000** (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.001*** (0.000)
Household size	-0.013 (0.020)	0.003 (0.054)	0.182*** (0.054)	0.114 (0.080)	-0.049 (0.065)	-0.094 (0.114)	0.121** (0.052)	-0.087 (0.076)
Number of children	0.030 (0.024)	0.012 (0.068)	-0.118* (0.062)	-0.082 (0.102)	0.098 (0.074)	0.211 (0.143)	-0.080 (0.058)	0.206** (0.095)
Employee	-0.005 (0.021)	0.030 (0.067)	0.174*** (0.058)	0.226** (0.102)	-0.035 (0.070)	0.046 (0.144)	0.081 (0.055)	0.225** (0.100)
Observations	858	298	748	259	740	259	740	259

Table VIII
Is Trust a Proxy for Risk Aversion?

The table shows regressions for the effect of trust on the number of stocks (Panel A) and on demand for health insurance (Panel B). The first panel reports Poisson count regressions for the number of different stocks in which the household invests, excluding equity in one's own company. The left-hand side variable is an integer varying between zero (no directly held stocks) and n (the household invests in n directly held stocks of different companies). In the last column the sample includes only households with strictly positive stockholdings. In Panel B the left-hand side is a dummy equal to one if the household has a private insurance. The reported figures are probit estimates calculated as the effect on the LHS of a marginal change in the RHS variable computed at the average value of the RHS variables. The last column shows Tobit estimates for the amount of health insurance purchased (i.e., the value of the premium paid). All characteristics are those of the household head. Standard errors are reported in parentheses. *** indicates the coefficient is different from zero at the 1% level, ** at the 5% level, and * at the 10% level.

Panel A. Trust and the Number of Stocks					
Trust	0.857*** (0.086)	0.863*** (0.086)	0.854*** (0.086)	0.754*** (0.087)	0.297*** (0.095)
Risk aversion	-0.013 (0.193)	0.071 (0.195)	0.060 (0.195)	-0.100 (0.198)	-0.171 (0.217)
Ambiguity aversion		-0.022** (0.009)	-0.022** (0.009)	-0.018** (0.008)	-0.024* (0.014)
Optimism		0.052 (0.046)	0.058 (0.046)	-0.064 (0.065)	
Own mutual funds			0.786*** (0.086)		
Financial wealth	0.006*** (0.000)	0.006*** (0.000)	0.006*** (0.000)	0.005*** (0.000)	0.002*** (0.000)
Income	-0.015 (0.012)	-0.016 (0.012)	-0.016 (0.012)	-0.025 (0.017)	-0.056** (0.028)
Male	1.002*** (0.141)	1.012*** (0.142)	1.017*** (0.141)	0.978*** (0.141)	0.965*** (0.161)
Age	-0.043*** (0.009)	-0.040*** (0.009)	-0.045*** (0.010)	-0.038*** (0.010)	0.005 (0.012)
Age squared	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.000 (0.000)
Household size	-1.015*** (0.098)	-1.025*** (0.099)	-1.031*** (0.099)	-1.085*** (0.101)	-0.934*** (0.105)
Number of children	1.479*** (0.117)	1.494*** (0.117)	1.503*** (0.118)	1.555*** (0.120)	1.178*** (0.122)
High school education	-0.672*** (0.104)	-0.648*** (0.104)	-0.666*** (0.105)	-0.629*** (0.106)	-0.890*** (0.109)
College education	-0.406*** (0.109)	-0.431*** (0.109)	-0.475*** (0.111)	-0.706*** (0.114)	-0.402*** (0.116)
Employee	0.257** (0.128)	0.268** (0.128)	0.255** (0.129)	0.227* (0.128)	0.178 (0.140)
Observations	1156	1156	1156	1156	162

(continued)

Table VIII—Continued

Panel B. Trust and the Demand for Health Insurance				
Trust	0.050 (0.031)	0.048 (0.031)	0.043 (0.031)	179.759 (223.050)
Risk aversion	-0.126 (0.079)	-0.137* (0.080)	-0.135* (0.080)	-773.808 (591.815)
Ambiguity aversion		0.000 (0.000)	0.000 (0.000)	0.188 (0.284)
Optimism			0.019 (0.016)	178.813 (115.943)
Financial wealth	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	5.767*** (1.411)
Income	1.867 (2.111)	1.852 (2.116)	1.797 (2.118)	6,981.391 (17,158.756)
Male	0.115*** (0.038)	0.115*** (0.038)	0.116*** (0.038)	750.441** (305.882)
Age	-0.011*** (0.003)	-0.011*** (0.003)	-0.013*** (0.004)	-34.931 (28.967)
Age squared	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.578* (0.334)
Household size	0.005 (0.037)	0.005 (0.037)	0.003 (0.037)	144.074 (269.719)
Number of children	0.002 (0.044)	0.002 (0.044)	0.005 (0.044)	-31.349 (320.463)
High school education	0.157*** (0.040)	0.156*** (0.040)	0.151*** (0.041)	451.967 (294.534)
College education	0.249*** (0.046)	0.249*** (0.046)	0.241*** (0.047)	986.477*** (318.290)
Employee	0.126*** (0.042)	0.126*** (0.042)	0.122*** (0.042)	71.651 (313.147)
Observations	1156	1156	1156	1156

Table IX
The Role of Personalized Trust

The table shows the effect of personalized trust on the participation in risky assets and the share invested in risky assets. Personalized trust is the trust a person has towards his bank official. In the first column the left-hand side variable is a dummy equal to one if the person invests in risky assets (directly held stocks, stock mutual funds, corporate bonds, derivatives); in the second and third is the share of financial wealth invested in these assets. "Risk averse" is a dummy variable equal to one if the interviewee answered (2) "Risk is an uncertain event from which one should seek protection" instead of (1) "Risk is an uncertain event from which one can extract a profit." All characteristics are those of the respondent. Standard errors are reported in parentheses. *** indicates the coefficient is different from zero at the 1% level, ** at the 5% level, and * at the 10% level.

	Probit for Ownership of Risky Assets	Share Invested in Risky Assets (Tobit Regression)	Conditional Share (Second Stage Heckman)
High personalized trust	0.1610*** (0.000)	0.0653*** (0.002)	0.0156 (0.280)
Medium personalized trust	0.0580 (0.121)	0.0226 (0.431)	0.0011 (0.955)
Averse to risk	-0.04* (0.025)	-0.0883*** (0.000)	-0.0730*** (0.000)
Financial wealth	0.0010*** (0.000)	0.0001*** (0.000)	0.00002*** (0.000)
Male	0.1050***	0.0753***	-
Age	0.0219***	0.0144***	0.0073***
Age squared	-0.0002***	-0.0001***	-0.00006***
Education	0.0221***	0.0138***	
Observations	1,834	1,834	1,834

- Panel A studies the number of stocks held.
- The trust coefficient is about 0.857 in the Poisson regression and remains large even after controlling for mutual-fund ownership.
- The paper interprets this as roughly 0.6 additional stocks for trusting households, which is nontrivial given a median of three stocks among stockholders.
- Panel B studies private health insurance. Trust has a positive sign there as well.

Why it matters. This is one of the paper's cleanest "horse race" tables. If trust were merely a proxy for low risk aversion, it should not raise both diversification and insurance demand.

5.7 Table IX: personalized trust in the banker

Read:

- High personalized trust in one's banker increases risky-asset ownership by about 0.161 in the Italian sample.
- This is economically large: the paper interprets it as about 16 percentage points, or roughly 25% of the sample mean.
- Personalized trust also increases the share invested in risky assets.

Why it matters. The trust effect is not confined to abstract generalized trust. It also shows up when trust is directed at the intermediary actually facilitating the transaction.

5.8 Table X and Figure 3: cross-country trust, participation, and ownership concentration

Read:

- Panel A of Table X regresses stock market participation on trust.

Table X
Trust, Stock Market Participation, and Ownership Concentration
around the World

Panel A shows the effect of trust on the share of stock market capitalization that is closely held and on the percent of the population that participates directly in the stock market in a cross-section of countries. Information on the fraction of stockholders across countries is obtained from Giannetti and Koskinen (2005), data on trust from the World Values Survey, and the remaining data from La Porta et al. (1998). Standard errors are reported in parentheses. *** indicates the coefficient is different from zero at the 1% level, ** at the 5% level, and * at the 10% level. Panel B shows the average participation rate when we divide the sample into four groups depending on whether the country was fully occupied during WWII and whether its level of trust is above or below the median (0.37).

Panel A:						
	% Population Participating in the Stock Market			% Stock Market Capitalization Closely Held		
	(1)	(2)	(3)	(4)	(5)	(6)
Trust (WVS)	0.272** (0.041)	0.399*** (0.001)	0.390*** (0.000)	-42.65** (0.023)	-46.80*** (0.01)	-46.84*** (0.01)
Legal enforcement		0.246*** (0.003)	0.143* (0.08)		-23.95* (0.074)	-21.68 (0.20)
Common law			0.091** (0.02)			-1.92 (0.82)
Observations	24	23	23	33	33	33
R-squared	0.18	0.50	0.62	0.15	0.24	0.25

Panel B: Level of Stock Market Participation		
	Country Occupied during WWII	Country Not Occupied during WWII
High trust (N)	0.19 (6)	0.19 (7)
Low trust (N)	0.09 (6)	0.19 (7)

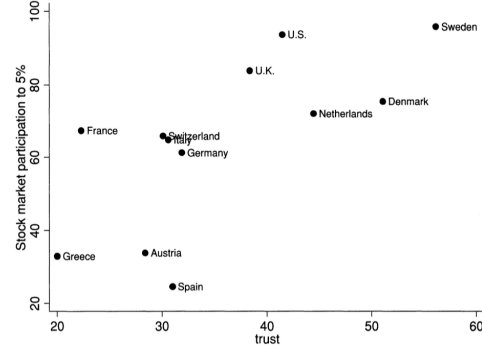


Figure 3. Trust and stock market participation of the wealthy. The figure plots stock market participation (direct and through mutual funds) for individuals in the top 5% of the wealth distribution in the countries in Table 1 against the average level of trust in that country (from the World Values Survey). The participation data are those shown in the second panel of Table 1, column "Top 5%" data. The data on trust are from the World Values Survey.

- The trust coefficient is positive and significant in all participation specifications (about 0.272, 0.399, and 0.390 depending on controls).
- For ownership concentration, the trust coefficient is strongly negative (about -42.65 , -46.80 , and -46.84).
- Figure 3 shows that among the top 5% wealth group, trust is strongly positively related to stock market participation.
- Panel B of Table X shows that trust matters especially in countries occupied during World War II: stock participation is about 19% in high-trust occupied countries versus about 9% in low-trust occupied countries, but there is essentially no difference in countries that were not occupied (19% versus 19%).

Why it matters. The cross-country evidence says trust affects not only who buys stock, but also how widely ownership is diffused in the economy.

6 Short summary

- The paper models trust as the subjective probability of not being cheated in financial transactions.
- Mistrust can rationally generate stock market nonparticipation even without fixed costs, and it strongly amplifies fixed participation costs when they are present.
- In Dutch household data, trust predicts:
 - direct stock market participation,
 - broader risky-asset participation,
 - the portfolio share invested in stocks and risky assets,
 - and the number of different stocks held.
- These results survive controls for risk aversion, ambiguity aversion, optimism, and expectations.

- Trust is stronger for the less educated, consistent with trust being a prior used when other information is weak.
- Trust also predicts private health insurance demand and risky-asset demand in Italian bank data, which helps separate it from risk tolerance.
- Across countries, more trust is associated with more stock market participation and less concentrated ownership.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that trust is an economically important determinant of stock market participation. The stock market is not just a risky-return environment; it is also a trust environment. If households attach even a small probability to being cheated, they may rationally stay out of the market or invest much less.

7.2 Economic intuition

The logic is simple but powerful:

- households cannot perfectly verify the honesty of firms and intermediaries,
- direct information transfer is difficult because parties with inside information have incentives to exaggerate,
- so investors use trust as a prior about whether the game is fair,
- and that prior matters most when they are less informed or less financially sophisticated.

This produces both micro and macro consequences:

- at the household level, lower participation and smaller risky portfolios,
- at the country level, thinner equity markets and more concentrated ownership.

7.3 Takeaway

This paper's lasting contribution is to show that limited stock market participation is not only about preferences over risk or about transaction costs. It is also about whether investors believe the market is trustworthy. If policymakers want broader participation, improving information and improving the perceived trustworthiness of financial institutions can matter as much as lowering explicit costs.